COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S4
DATE PERFORMED:	8/28/2026	**DATE SUBMITTED:**	
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ACTIVITY NO. 6 SEARCHING TECHNIQUES

1. PROCEDURE OUTPUT

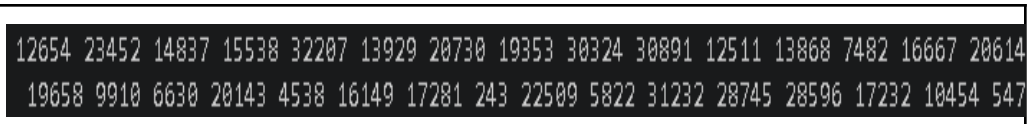
Screenshot	
Observations	The data that was generated from the dataset. Every time we run the code, it will generate a new random set of data.

Table 6-1. Data Generated and Observations.

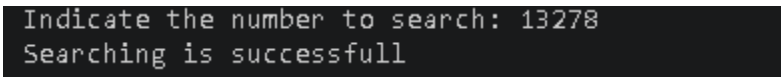
Code	
Output	
Observations	The idea of this function is to search for data sequentially that was generated by the dataset array. If the user inputs the write data, it will print successful and if not, it will be unsuccessful.

Table 6-2a. Linear Search for Arrays

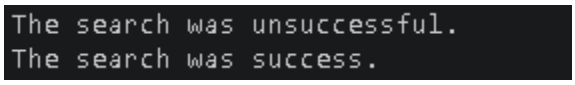
Code	
Output	
Observations	This function is a linear search for linked list where we implement nodes with a type of character variable. The idea of this function is to search if the data exist in the nodes. It will search for a specific node that have the data that we want to find in sequence starting from name1 and so on until it finds the data that we want, and of course if we want to find a data that doesn't exists, the function will go through all of the nodes until reaches nullptr and prints out search unsuccessful.

Table 6-2b. Linear Search for Linked List

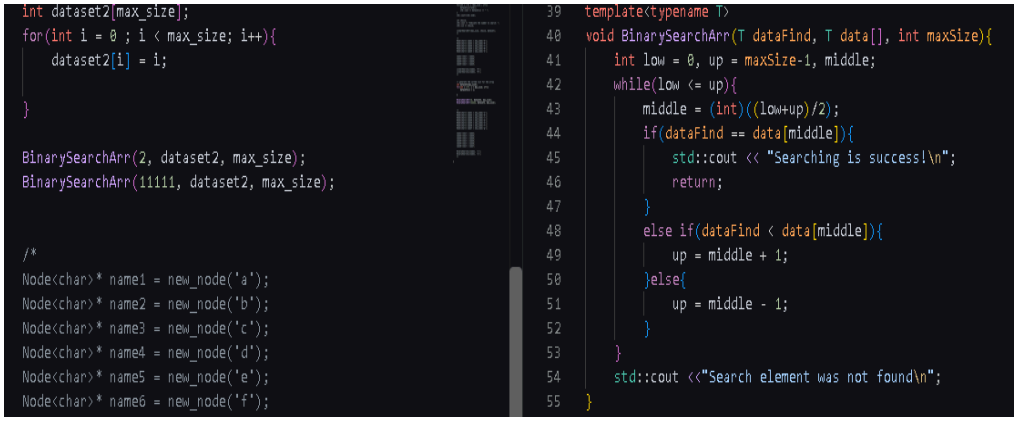
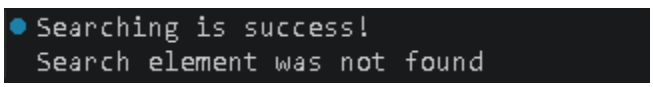
Code	
Output	
Observations	This function searches data if it is equal to middle value/data. For example we input 2, then the function will add 2 + 0 = 2 and divided it by 2 = 1. This means the 1 value/data is the middle and if the condition is if the middle is equal to the data that we found, then it will be successful. If the dataFind is less than the middle, then the function will add 1 to be equal, and if the dataFind has a bigger value than the middle data, the function will subtract 1 to be equal to the middle data.

Table 6-3a. Binary Search for Arrays

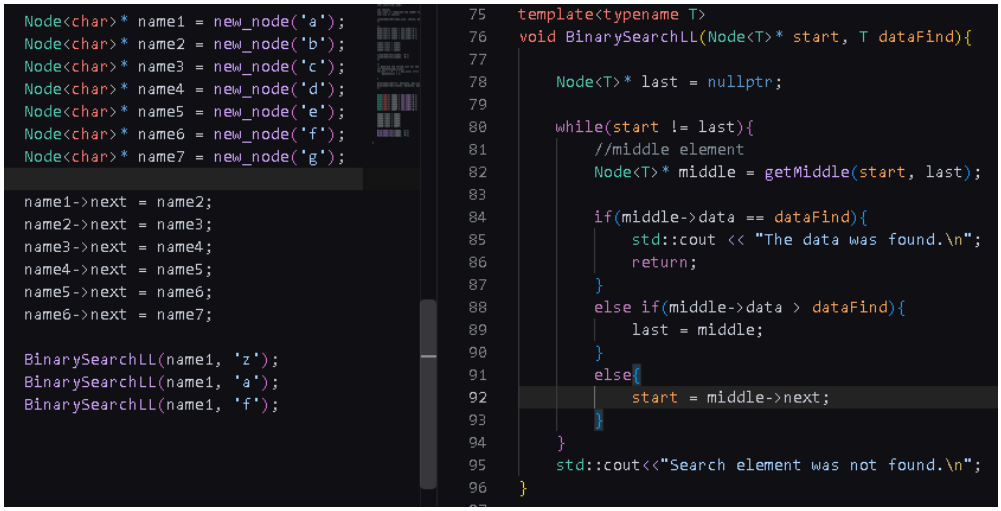
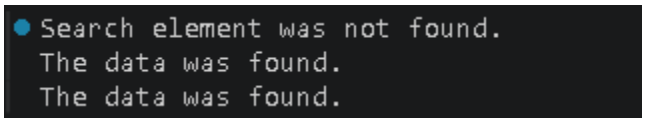
Code	
Output	
Observations	The idea of this is to search data in binary in a linked list. This function is almost the same as the linear search linked list function where the function will go through all of the nodes and check each node if the data that we want to find is in that node. The getMiddle function searches data in slow and fast pace, that's why it is called binary search because it is a lot faster to find the data and it double checks it in slow pace.

Table 6-3b. Binary Search for Linked List

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or

completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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